Section 5.5 Multiple Eigen Value Solutions

In Section 5.3 we stated that when we have a repeated eigen value of a matrix there may or may not be more than one eigen vector associated with it.

Let's consider a system of two linear first order equations of the form $\vec{x}' = A\vec{x}$

And let's suppose that we have a repeated eigen value of λ that only has one eigen vector associated with it $\vec{v}_1$

We said that in this case the solution to the system is $\vec{x} = c_1\vec{v}_1 e^{\lambda t} + c_2\left(\vec{v}_1 t + \vec{v}_2\right)e^{\lambda t}$

Where $\vec{v}_2$ is called a generalized eigen vector. Let's see how we might find this eigen vector by plugging it in to the differential equation itself

So for the $\vec{x}(t)$ above to be a solution the generalized eigen vector $\vec{v}_2$ must satisify $(A - \lambda I)\vec{v}_2 = \vec{v}_1$

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Let's consider the system: $\begin{bmatrix} x_1' \\ x_2' \end{bmatrix} = -\begin{bmatrix} 1 & -3 \\ 3 & 7 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$ and find the general solution and the generalized eigen vector.

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Let's just consider the general case of a repeated eigen value for a matrix A. We know that the way we find eigen values is to solve the characteristic polynomial coming from the equation

$$\det(A - \lambda I) = 0$$

For an nxn matrix A we know we could factor that equation into a product of linear terms involving the eigen values.

$$(\lambda - \lambda_1)(\lambda - \lambda_2)\cdots(\lambda - \lambda_n) = 0$$

If we have a repeated eigen value this might look more like

$$(\lambda - \lambda_1)^m(\lambda - \lambda_2)\cdots(\lambda - \lambda_{n-k}) = 0$$

Here we would say that the eigen value λ_1 whould have algebraic multiplicity m. If when solving for the eigen vectors for λ_1 we find m linearly independent ones we say that λ_1 is **complete.** If there are only $p < m$ linearly independent eigen vectors we call it a **defective** eigen value. Further we call the number of missing eigen vectors $d = m - p$ the **defect.** This will also be the number of generalized eigen vectors that we can find.

We can find all of the generalized eigen vectors belonging to a defective eigen value successively. Suppose that $\vec{v}_1$ is one of the regular eigen vectors of λ, A which means that it satisfies.

$$(A - \lambda I)\vec{v}_1 = 0$$

From what we saw before a generalized eigen vector can be found solving

$$(A - \lambda I)\vec{v}_2 = \vec{v}_1$$

for $\vec{v}_2$. The next generalized eigen vector may be found with

$$(A - \lambda I)\vec{v}_3 = \vec{v}_2$$

All the way up to how ever many generalized eigen vectors might be generated from $\vec{v}_1$

$$\vdots$$
$$(A - \lambda I)\vec{v}_k = \vec{v}_{k-1}$$

Note the following pattern using the above sequence:

$$(A - \lambda I(A - \lambda I)\vec{v}_2 = (A - \lambda I)\vec{v}_1 = 0 \;\rightarrow\; (A - \lambda I)^2\vec{v}_2 = 0$$

then

$$(A - \lambda I)^2(A - \lambda I)\vec{v}_3 = (A - \lambda I)^2\vec{v}_2 = 0 \rightarrow (A - \lambda I)^3\vec{v}_3 = 0$$

And so on until

$$(A - \lambda I)^k\vec{v}_k = 0$$

Here we would say that k us the rank of the generalized eigen vector.

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We would call the set of vectors we found $\{\vec{v}_1, \vec{v}_2, \dots, \vec{v}_k\}$ a length k chain of generalized eigen vectors based on the eigen vector $\vec{v}_1$.

It may not be that we get all of our generalized eigen vectors from $\vec{v}_1$ however. If we have other regular eigen vectors coming from lambda they may also be generators of some of the generalized eigen vectors of λ and A.

Let's choose a different notation for generalized eigen vectors now, let's denote them with a u and the regular eigen vectors with a v.

To illustrate the possibilities with a defective eigen value and its generalized eigen vectors let's suppose the matrix A has an eigen value λ with multiplicity $m = 4$ and defect $d = \cdots$.

$d = 0$, Then we have $\{\vec{v}_1\}, \{\vec{v}_2\}, \{\vec{v}_3\}, \{\vec{v}_4\}$ four length 1 chains made up of regular eigen vectors

$d = 1$, Then we have $\{\vec{v}_1, \vec{u}_1\}, \{\vec{v}_2\}, \{\vec{v}_3\}$ two length 1 chains of regular eigen vectors and a length 2 chain of a regular eigen vector and a generalized eigen vector coming from it.

$d = 2$ Then we can have $\{\vec{v}_1, \vec{u}_1\}, \{\vec{v}_2, \vec{u}_2\}$ two length two chains with a regular eigen vector and a generalized eigen vector generated from each.

$d = 2$ We could also have $\{\vec{v}_1, \vec{u}_1, \vec{u}_2\}, \{\vec{v}_2\}$ One length 3 chain and one length 1 chain.

$d = 3$ $\{\vec{v}_1, \vec{u}_1, \vec{u}_2, \vec{u}_3\}$ a length 4 chain with generalized eigen vectors all coming from the only regular one found.

Suppose we indeed have an eigen value λ with multiplicity 4 and defect 2, how can we know weather the vectors we need come in the form $\{\vec{v}_1, \vec{u}_1\}, \{\vec{v}_2, \vec{u}_2\}$ *or* $\{\vec{v}_1, \vec{u}_1, \vec{u}_2\}, \{\vec{v}_2\}$ Well we will be able to know but first let's give an aloorithim for constructing the chains… Also head here for a nice set of notes going through two examples like this… https://www.math.purdue.edu/~neptamin/303Au21/Handouts/High_defect.pdf

We notice that in every case above the length of the chain was at most $d + 1$ so we could begin looking for generalized eigen vectors by solving

$$(A - \lambda I)^{d+1} \vec{u}_{d+1} = 0$$
and
$$(A - \lambda I)^{d} \vec{u}_{d+1} \neq 0$$

If we get a solution call it $\vec{u}_{d+1}$ then since we have the relationship $(A - \lambda I)\vec{u}_{m+1} = \vec{u}_m$ we can multiply $\vec{u}_{d+1}$ by $A - \lambda I$ to get the next generalized eigen vector $\vec{u}_d$. Then we can do the same thing to $\vec{u}_d$ to get $\vec{u}_{d-1}$ all the way down until we get $(A - \lambda I)\vec{u}_1 = 0$ at which point we know we have hit a regular eigen vector as we got zero so that $\vec{u}_1$ is a regular eigen vector $\vec{v}_1$

(1) Try to solve $(A - \lambda I)^{d+1} \vec{u}_{d+1} = 0$, $(A - \lambda I)^{d} \vec{u}_{d+1} \neq 0$ if you get a solution then do
(2) $(A - \lambda I)\, \vec{u}_{d+1} = \vec{u}_d$
$(A - \lambda I)\, \vec{u}_d = \vec{u}_{d-1}$
$\vdots$
$(A - \lambda I)\, \vec{u}_2 = \vec{u}_1$
$(A - \lambda I)\, \vec{u}_1 = 0$
(3) once you get to zero the set $\{\vec{u}_1, \vec{u}_2, \dots, \vec{u}_{d+1}\}$ is the chain for λ with $\vec{u}_1$ being a regular eigen vector of A.
If you cannot solve the equations in (1) move down from $d + 1$ to d, if you can't solve it with d step down until you can.

This last line in the algorithm is how we can tell between a situation like $\{\vec{v}_1, \vec{u}_1\}, \{\vec{v}_2, \vec{u}_2\}$ *or* $\{\vec{v}_1, \vec{u}_1, \vec{u}_2\}, \{\vec{v}_2\}$ If we started with $d + 1 = 3$ and we get a solution to the equation in step (1) we would be in the situation $\{\vec{v}_1, \vec{u}_1, \vec{u}_2\}, \{\vec{v}_2\}$. If not we would step down to d and solve $(A - \lambda I)^2 \vec{u}_2 = 0$ and we then know we would be in situation $\{\vec{v}_1, \vec{u}_1\}, \{\vec{v}_2, \vec{u}_2\}$ and we would get two different solutions to form the chains from .

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We know we will always be able to build a solution with the needed number of linearly independent vectors because of these two facts from linear algebra

(1) Any chain of generalized eigen vectors will be a linearly independent set of vectors
(2) If two chains of generalized eigen vectors are based on linearly independent eigen vectors then their union will be a linearly independent set of vectors.

So if we get $a\ k\ chain\ \{\vec{u}_1, \vec{u}_2, \ldots \vec{u}_k\}$ of eigen vectors we will have a contribution to the full general solution of the linear system of k functions of the form

$$\vec{x}_1 = \vec{u}_1 e^{\lambda t}$$
$$\vec{x}_2 = (\vec{u}_1 t + \vec{u}_2) e^{\lambda t}$$
$$\vec{x}_3 = (\frac{1}{2}\vec{u}_1 t^2 + \vec{u}_2 t + \vec{u}_3) e^{\lambda t}$$
$$\vdots$$
$$\vec{x}_k = \left(\frac{1}{(k-1)!}\vec{u}_1 t^{k-1} + \frac{1}{(k-2)!}\vec{u}_2 t^{k-2} + \frac{1}{(k-3)!}\vec{u}_3 t^{k-3} + \cdots \vec{u}_{k-1} t + \vec{u}_k\right) e^{\lambda t}$$

So that $\vec{x}(t) = other\ parts\ from\ other\ eigen\ vals + c_1\vec{x}_1 + c_2\vec{x}_2 + \cdots + c_k\vec{x}_k$

What would be the structure of the contributions of the eigen value with multiplicity 4 and defect 2 for the two possible situations of $\{\vec{v}_1, \vec{u}_1\}, \{\vec{v}_2, \vec{u}_2\}$ or $\{\vec{v}_1, \vec{u}_1, \vec{u}_2\}, \{\vec{v}_2\}$?

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Let's do an example with the help of octave

$$\vec{x}' = \begin{bmatrix} 0 & 1 & 2 \\ -5 & -3 & -7 \\ 1 & 0 & 0 \end{bmatrix} \vec{x}$$